5 Security and privacy

This chapter covers

- Understanding basics of cryptography and public-private key pairs
- Managing digital identity for decentralized participants using public-key cryptography
- Using cryptography and hashing for the privacy and security of blockchain data
- Illustrating security and privacy concepts using blind auction smart contract
- Deploying smart contracts on a public blockchain

Security and privacy are concerns in any system open to public access, from public buildings and highways to hardware and software systems. But they are especially serious concerns in blockchain-based systems. These systems operate beyond traditional boundaries of trust, such as the one established by a medical provider for its patients or by a university for its enrolled students. Security in these systems is typically established by verifying government-issued credentials, such as a driver's license and passport, authentication using usernames and passwords, and end-to-end encryption of messages and communications.

Advances in the digitization of health information, student records, and the like have led to regulations to ensure data privacy for the participants. The Health Insurance Portability and Accountability Act of 1996 (HIPAA) in the United States provides data privacy and security provisions for safeguarding medical information. The Family Educational Rights and Privacy Act of 1974 (FERPA) is a federal law that protects the privacy of student education records. The data records in these systems are typically housed in a centralized database, and its